Adaptive Traffic Signal Optimization Using Reinforcement Learning and YOLOv8

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***Abstract*—Urban traffic congestion is a growing challenge due to increasing vehicle density and the limitations of conventional fixed-time traffic signal systems. These systems lack adaptability and fail to respond effectively to real-time traffic variations, especially in heterogeneous traffic environments such as Indian roads. This paper proposes an intelligent traffic signal control framework that integrates Deep Reinforcement Learning (DRL) with computer vision for adaptive signal optimization. The system employs Proximal Policy Optimization (PPO) as the primary learning algorithm, with comparisons against Deep Q-Network (DQN) and Advantage Actor-Critic (A2C). Traffic conditions are simulated using the SUMO traffic simulator, controlled in real time via the TraCI interface. Vehicle density and congestion states are extracted using YOLOv8-based perception from rendered simulation frames. Experimental results demonstrate significant reductions in average waiting time and queue length compared to fixed-time controllers, highlighting the effectiveness of the proposed approach.**

***Index Terms*—Traffic Signal Control, Reinforcement Learning, PPO, SUMO, TraCI, YOLOv8, Intelligent Transportation Systems**

I. Introduction

Rapid urbanization has resulted in increased traffic congestion, longer travel times, fuel wastage, and higher emissions. Traditional traffic signal systems operate using predefined fixed-time schedules that do not adapt to real-time traffic conditions. Such approaches are inefficient in dynamic and heterogeneous traffic scenarios. Recent advancements in Artificial Intelligence (AI), particularly Reinforcement Learning (RL), offer promising solutions for adaptive traffic signal control.

This research presents an AI-driven traffic signal optimization system that integrates Deep Reinforcement Learning with computer vision. By leveraging real-time vehicle count information and learning optimal signal timing strategies, the proposed system aims to reduce congestion and improve overall traffic efficiency.

II. Problem Statement

Conventional traffic signal systems suffer from fixed signal timing mechanisms that fail to adapt to fluctuating traffic demands. These systems lead to excessive waiting times, inefficient road utilization, increased fuel consumption, and elevated pollution levels. There is a lack of intelligent decision-making and learning capability in existing signal controllers, particularly in complex urban traffic environments.

III. Motivation

The motivation behind this work stems from the growing inefficiency of static traffic signal systems and the increasing availability of simulation tools and AI techniques. Reinforcement Learning enables traffic controllers to learn optimal strategies through interaction with the environment, while computer vision provides accurate perception of real-time traffic states.

IV. Literature Review

Li et al. [1] demonstrated deep reinforcement learning for adaptive traffic signal timing. Wei et al. [2] proposed IntelliLight, a DRL-based framework for traffic signal control. Arel et al. [3] introduced multi-agent reinforcement learning for network-level traffic optimization. Van der Pol and Oliehoek [4] explored coordinated deep RL approaches for traffic light control. However, most existing works rely on simulator-based state representations and lack vision-based perception.

V. Research Gap

Most existing traffic signal control approaches depend on predefined simulator states and do not incorporate real-time visual perception. Additionally, limited work explores PPO-based optimization with detailed training analysis using TensorBoard. This work bridges these gaps by integrating YOLOv8 perception, PPO learning, and comprehensive experimental evaluation.

VI. Objectives

- Develop an adaptive traffic signal control system using PPO.
- Integrate YOLOv8 for real-time vehicle detection.
- Compare PPO with DQN and A2C algorithms.
- Reduce average waiting time and queue length.

VII. Proposed Methodology

A. System Architecture

Fig. 1 illustrates the complete system architecture integrating SUMO, YOLOv8, and PPO.

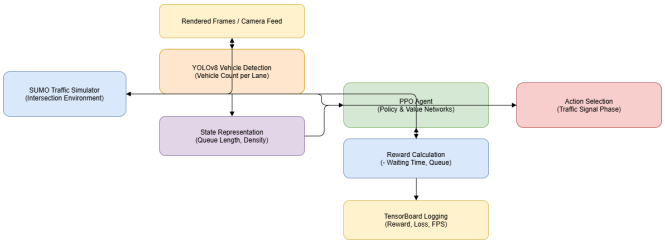

Fig. 1. Architecture of the Proposed YOLOv8 + PPO Based Traffic Signal Control System

B. Reinforcement Learning Formulation

State: Vehicle count, queue length, and lane density
Action: Traffic signal phase selection
Reward: Negative weighted sum of waiting time and queue length

VIII. Algorithm

Algorithm 1 PPO-Based Traffic Signal Control

```
Initialize policy and value networks
for each episode do
    Reset SUMO environment
    for each timestep do
        Observe state s_t
        Select action a_t
        Apply action via TraCI
        Receive reward r_t
    end for
    Update PPO networks
end for
```

IX. Experimental Setup

Experiments were conducted using a four-way intersection in SUMO. PPO, DQN, and A2C were implemented using Stable-Baselines3. Training performance was monitored using TensorBoard.

X. Results and Discussion

A. Training Performance

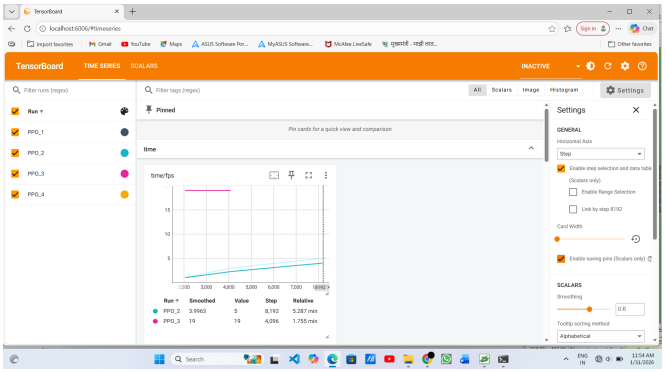

Fig. 2. TensorBoard FPS Performance During Training

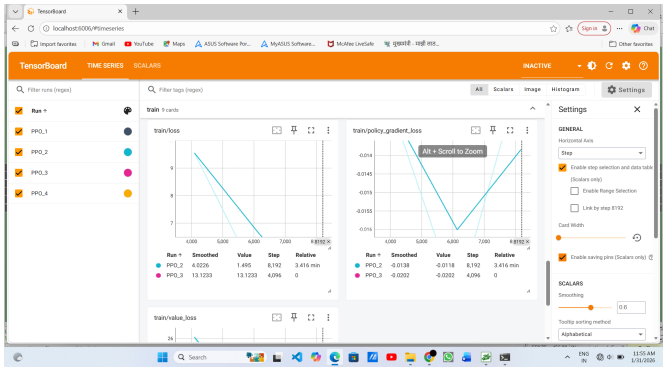

Fig. 3. TensorBoard Training Loss Metrics

B. Simulation Output

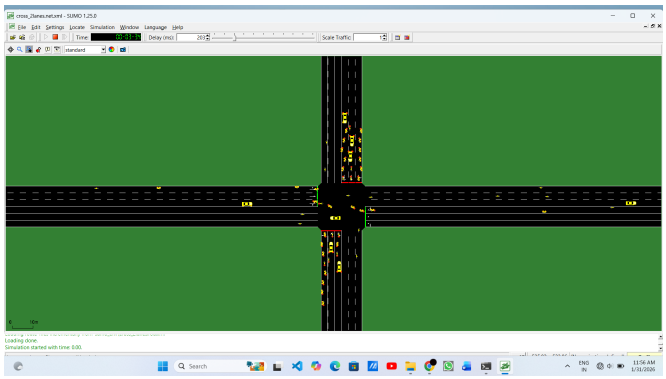

Fig. 4. SUMO Simulation Output with Adaptive Traffic Signal Control

C. Quantitative Results

TABLE I
Performance Comparison

Method	Waiting Time (s)	Queue Length	Throughput
Fixed-Time	78.5	24.3	Low
DQN	52.1	16.8	Medium
A2C	48.6	15.2	Medium
PPO	**35.4**	**10.6**	High

XI. Implementation and Reproducibility

The complete implementation and experimental code are publicly available at:

https://github.com/rupalikale15/Traffic_Signal_Optimization_using_Reinforcement_Learning

XII. Conclusion

This paper presented an intelligent traffic signal control system integrating YOLOv8-based perception with PPO reinforcement learning. The proposed approach significantly reduced congestion and waiting times compared to traditional methods. Future work will extend this framework to multi-intersection networks and real-world deployments.

REFERENCES

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